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Improvement of Brain Tumor Categorization using Deep Learning: A Comprehensive Investigation and Comparative Analysis

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Abstract

A brain tumor is a critically severe health disorder that requires an accurate and timely diagnosis for effective treatment. Advances in medical imaging and deep learning methods have shown potential for enhancing the identification and categorization of brain cancers throughout the years. In the present research, our study compares the accuracy of eight different deep learning models in the classification of brain tumors employing brain MRI data that involve Densenet121, EfficientNet B7, InceptionResNetV2, Inception_V3, RestNet50V2, VGG16, VGG19, and Xception. To further improve performance, we propose integrating a hybrid deep learning technique. Efficient and timely diagnosis of brain tumors is critical for the treatment of patients, and our study aims to achieve high recall, accuracy, and F1-score in this context. With a precision of 96.63%, our innovative convolutional neural network (CNN) technique achieved outstanding results in brain tumor diagnosis. Also, our study investigates the unique capabilities of certain models, such as VGG19 and InceptionResNetV2, and their possibilities for better glioma tumor detection efficiency. Our results, in particular, provide insight into possible uses of deep learning frameworks, including the integration of hybrid techniques, in medical imaging, offering an innovative approach for increased brain tumor detection and identification.

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Keywords: Brain Tumor, Deep Learning Algorithms, Medical Imaging, Image Classification, Neural Network Models;

1. Introduction

Cancer is one of the most widespread illnesses on the globe, with an estimated 1.8 million new diagnoses and 600,000 deaths in the US in 2020. Cancer is characterized by the unregulated development of abnormal cells within the body. It arises from changes in cellular function or defects that make it more difficult for a cell to undergo controlled cell death [1]. This affects many tissues and organs and promotes the growth of malignancies. The growth can push on brain tissue, resulting in serious problems. even if the tumor is harmless. Brain tumor cause about 2.17% of all cancer deaths,

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and the five-year survival rate for glioblastoma is weak, at roughly 5.6%. The incidence of brain tumor and the worrying statistics have driven ongoing study on the issue, with doctors and scientists seeking strategies to avoid tumor, create more effective medications, develop better diagnostic tools, and develop better methodologies to assess and categorize tumor. This study incorporates novel approaches for investigating brain architecture as well as the creation of AI technologies [3].

In recent years, the merging of AI technologies and healthcare imaging has shown significant potential for enhancing the precision and efficacy of diagnosing a variety of medical disorders. The detection and classification of brain tumor in magnetic resonance imaging (MRI) images is a critical focus. The efficiency of neural networks, the advantages of transfer learning for healthcare imaging, the impact of data augmentation on the efficiency of models, and the possible benefits of the cross-transformer network will all be thoroughly discussed.

2. Literature review

The realm of brain tumor analysis via MRI imaging has been extensively explored in recent studies. Kumar et al. [4] conducted a meticulous investigation into segmentation techniques, revealing discrepancies in accuracy and efficiency across methods. Advocating for hybrid segmentation involves combining the strengths of different methods and adapting methodologies to accommodate diverse tumor types and imaging scenarios. Liu et al. [5] introduced transformer-based approaches to medical image analysis, improving diagnostic precision, though challenges inadapting transformers and managing computational intensity were acknowledged. ZainEldin H et al. [6] proposed a novel technique using deep learning and optimization, showing promise in detection and classification, but dataset diversity and generalization challenges arose. Amin, J. et al.'s [7] survey highlighted machine learning methods' promise in brain tumor detection and classification, suggesting larger, diverse datasets and advanced algorithms to address data imbalance and generalization issues. Anagun et al. [8] evolved a brain tumor detection method that utilizesCNN with promising accuracy, while emphasizing dataset diversity, class imbalance, and interpretability. Integrating insights from these studies reveals the evolving landscape of brain tumor analysis, calling for hybrid methods, domain-specific strategies, diverse datasets, optimization refinement, and explainability techniques to enhance accuracy and patient care. Studies by Badza and Barjaktarovic [9], Khan et al. [11], and Hassan Ali Khan et al. [13] collectively highlight the limitations posed by small datasets, emphasizing data augmentation as a pivotal tool to augment sample diversity and combat overfitting. A consistent theme across Srinivasan et al. [10] and Khan et al. [11] is the imperativefor dataset diversity and the mitigation of biases, emphasizing the need for larger, inclusive datasets and techniques toensure equitable model performance. The optimization of CNN architectures and hyperparameters, as highlighted by Badza and Barjaktarovic [9] and Hassan Ali Khan et al. [10], emerges as a central avenue for enhancing model accuracyand generalizability. Furthermore, transfer learning, advocated by Srinivasan et al. [10] and Khan et al. [11], offers a potent means to leverage pre-existing knowledge and improve adaptability when confronting different tumor types andimaging modalities. The challenge of interpretability and explainability, articulated by Masood et al. [12], underscoresthe need for tailored solutions that bridge the gap between complex model outputs and clinical understanding. In synthesis, an integrative approach that leverages data augmentation, optimizes architectures, promotes diversity, ensures fairness, explores transfer learning, and enhances interpretability holds the potential to propel the accuracy andclinical relevance of DL models in brain tumor categorization. Collective efforts towards enriched datasets and interpretable AI methodologies are pivotal in driving advancements in this critical medical domain.

The realm of brain tumor identification and classification has been significantly propelled by the integration of DL and ML approaches, as evidenced by a series of noteworthy studies. Biratu et al. [14] conducted an encompassing survey of prevailing algorithms in this domain, unveiling challenges centered on precision and resilience. To surmount these obstacles, the authors recommended the amalgamation of advanced machine learning methods and the utilization of more expansive and diverse datasets, a consensus echoed by subsequent research. Mahmud et al. [15] pioneered deep learning networks for brain tumor detection, achieving heightened accuracy, though constrained datasets and potential overfitting surfaced as hurdles. This study underscored the need for partnerships with medical institutions to access larger datasets and the application of data augmentation and transfer learning to enhance generalization. Tiwari et al. [16] harnessed CNN for multiclass brain tumor detection, yielding encouraging accuracy. However, challenges in accommodating tumor characteristic variations and image quality fluctuations were acknowledged, with solutions

3. Methodology

This study presents a method for categorized brain tumor using multiple deep learning frameworks such as Densenet121, EfficientNet B7, InceptionResNetV2, Inception_V3, RestNet50V2, VGG16, VGG19, Xception, and a custom CNN. There are three categorization modules to examine: "BTD","MRI-D," and "Augmented dataset." In both the training and validation sets, the dataset is rigorously balanced among classes. The models display high feature extraction skills after being pretrained on ImageNet. Accuracy, F1 score, sensitivity, specificity, and precision are typical measures used to evaluate efficiency [8].

3.1. Dataset

Three datasets have been employed in this research: The Brain Tumor Dataset (BTD), the MRI-D dataset, and an augmented dataset.

- BTD Dataset

The BTD dataset is sourced from Kaggle Table 1 shows the split of images in the training and validation categories for both the "No Tumor" and "Tumor" groups, as well as the overall number of images within each folder. Sample image of BTD Dataset and MRI-Dataset is shown in figure 1.

Table 1. Tumor and No tumor BTD Dataset.

Label	No Tumor	Tumor	Total Images
Training Folder	1450	1450	2900
Validation Folder	1450	1450	2900

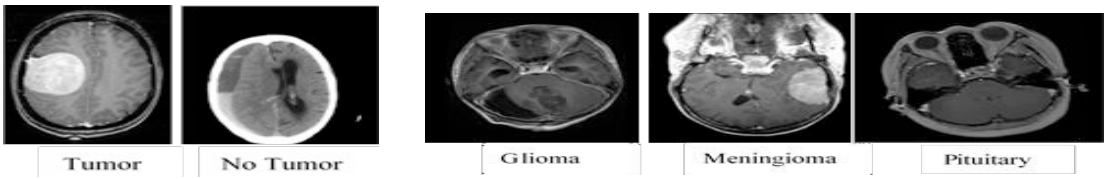


Fig. 1. Sample image of BTD Dataset and MRI-Dataset.

- MRI-Dataset

The number of validation and training images for each of the three classes is shown in Table 2. Each class has 1000 training and 1000 validation images, for an aggregate of 2000 images.

Table 2. Glioma, meningioma and Pituitary MRI-Dataset.

Label	Glioma	Meningioma	Pituitary	Total Images
Training Folder	1000	1000	1000	3000
Validation Folder	1000	1000	1000	3000

- Augmented dataset

Combining the MRI-D and BTD datasets, the augmented dataset is formed. The arrangement of images among different folders in the brain tumor dataset is as follows in table 3: The folder names like “Glioma", "Meningioma”, “No Tumor", and "Pituitary" comprised 1000 images each. The "Training Folder" contains 1000 images that are equally spread out among the three tumor and one no tumor types. The "Validation Folder" contains 4000 images, which replicates the arrangement of the training data, increasing the entire number of images in the dataset by over 4000 The "Testing Folder" contains 300 images, which replicates the arrangement of the testing data, these further data for each class are used for making confusion matrixes for all eight deep learning models.

Table 3. Augmented dataset.

Label	Glioma	Meningioma	No_Tumor	Pituitary	Total Images
Training Folder	1000	1000	1000	1000	4000
Validation Folder	1000	1000	1000	1000	4000
Testing Folder	300	300	300	300	1200

3.2. CNN Architecture

Convolutional neural networks (CNNs) are essential for image-based activities, especially medical imaging applications like classifying brain tumors. To comprehend and classify intricate patterns in medical images, the architecture depicted in the figure is intended to extract hierarchical characteristics from 3-channel MRI scans. CNN Architecture is shown in figure 2.

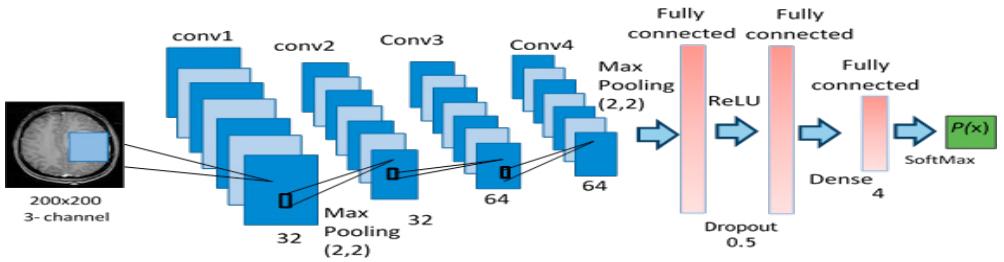


Fig. 2. CNN Architecture.

3.3. DL techniques for brain tumor categorization

The presented methodology uses an advanced CNN-based method for classifying brain tumors, starting with the acquisition of brain MRIs, which are the main diagnostic method for tumors [11]. MRI images are scaled and cropped in order to provide consistent input. Then, using transfer learning rather than building a CNN from begin, data is carefully divided into separate training, validation, and future test sets [12]. Major pre-trained models, such as DenseNet 121, EfficientNet B7, and InceptionV3, which were trained on large-scale datasets like ImageNet, are used to extract features from MRI scans by identifying important patterns associated with tumor. The structure is expanded by adding layers after feature extraction. A probability distribution across the different tumor classifications is provided by the concluding layer, which has a Softmax activation and enables precise classification of tumor types such gliomas, meningiomas, and pituitary [13]. A variety of measures are used in the evaluation of algorithmic performance, including as precision, recall, accuracy, and F1-score after prediction. Deep Learning-Based Brain Tumor Classification Process is shown in figure 3.

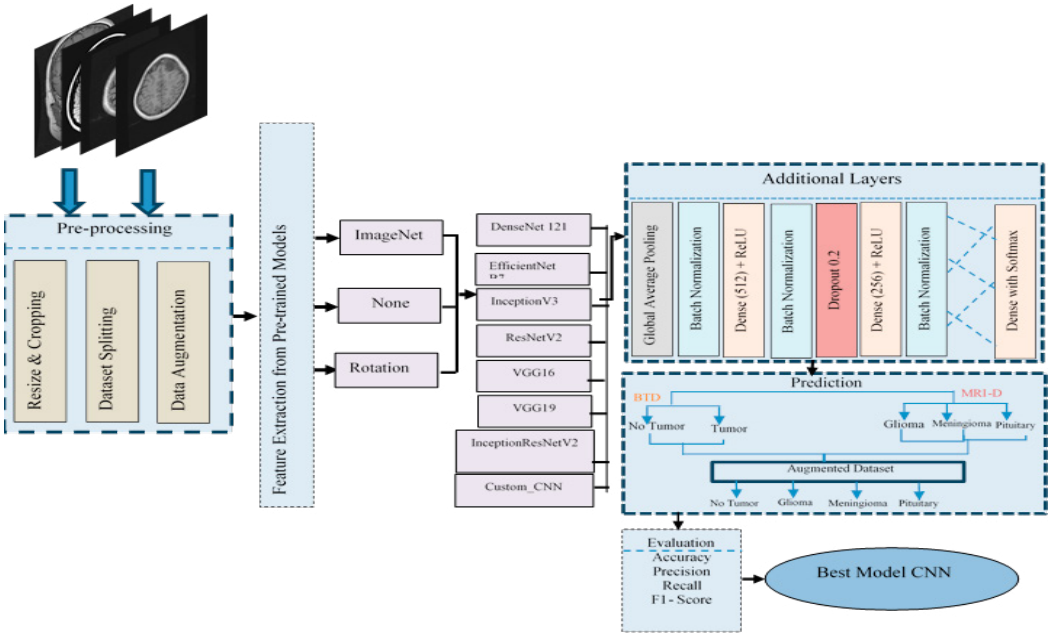


Fig. 3. Deep Learning-Based Brain Tumor Classification Process.

3.4. Performance evaluation

The hyperparameters for the augmented dataset's brain tumor classification are listed in Table 4. For effective model training, a batch size of 16 was used, processing 16 data in each iteration. The algorithm was able to iteratively optimize its parameters using the increased brain tumor dataset over the period of a 10-epoch training process. The output layer's activation function, Softmax, generated probability distributions for different tumor types. The loss measure, which was appropriate for the multi-class classification challenge, was categorical crossentropy, and the Adam optimizer dynamically modified learning rates to improve convergence. In the case of brain tumor classification, these uniform hyperparameters among various algorithms promoted a common training framework that allowed for insightful comparisons.

Table 4: Hyperparameters for Augmented dataset

Batch size	16
Epoch	10
Activation Function	Softmax
Optimizer	Adam
Loss Function	Categorical crossentropy

Our employ a number of fundamental measures [15], to assess the performance of our brain tumor categorization method. These parameters include accuracy, F1 score, recall and precision.

4. Purpose

The goal of this study is to improve medical imaging, particularly the use of MRI data for brain tumor categorization. A thorough assessment of eight deep learning models with a focus on accuracy, recall, precision, and F1-score is the key objective. The detection of glioma tumors is critical for prompt and efficient treatment, and this is the main emphasis of the study. By presenting a novel Convolutional Neural Network (CNN) technique, the study aims to set a new benchmark for high-performance brain tumor detection. Understanding the performance of models like VGG19 and InceptionResNetV2 is aimed at expanding the knowledge base of deep learning frameworks in medical imaging. Improving the detection of brain tumors is the ultimate aim, with potential applications for better patient outcomes and medical treatments.

5.Major Research Findings

5.1. BTD Dataset

a) Evaluation of Pre-trained DL Algorithms with MRI-D Dataset

Figure 4 depicts how pre-trained models such as DenseNet121, InceptionResNetV2, and VGG16 perform in binary tumor categorization as a result of exposure to varied ImageNet data. Due to a lack of exposure, models such as EfficientNetB7 and InceptionV3, particularly without pre-training, perform poorly. These findings highlight the critical significance of pre-training in the acquisition of useful picture attributes. Even when not pre-trained, EfficientNetB7 and InceptionV3 demonstrate potential, indicating their linked capabilities. The graph emphasizes the need for pre-training for higher tumor classification accuracy, with several model selections based on the dataset and training circumstances.

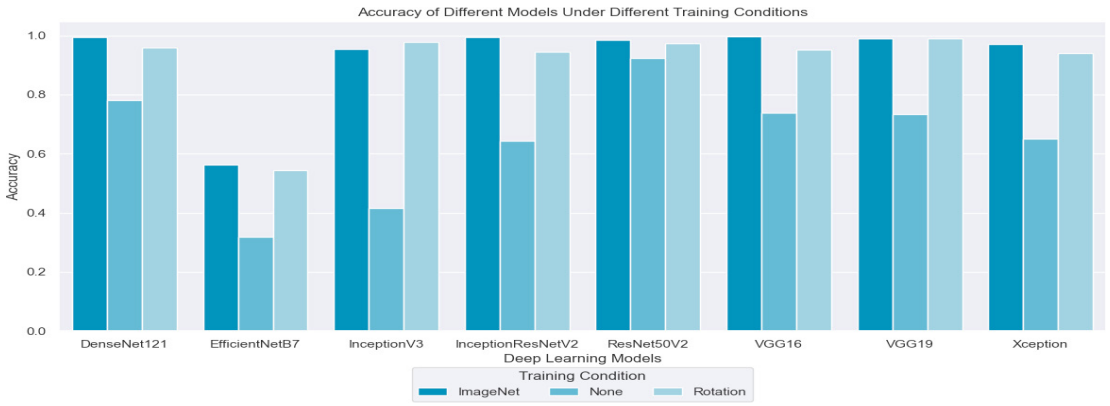


Fig. 4. Evaluation of Pre-trained DL Algorithms for Binary Tumor Categorization.

4.1. MRI Dataset

a) DL Framework Effectiveness for Brain Tumor Identification

Xception with ImageNet weights is the most effective approach shown in Figure 5, continuously outperforming on the basis of mean precision throughout every case. Its exceptional and constant performance outperforms all other designs, making it the best option for brain tumor categorization. VGG16 and VGG19, both initiated with ImageNet weights, show robust and reliable efficiency, frequently acquiring high mean accuracies. This constancy highlights their aptitude for identifying brain cancers. When tested across situations, EfficientNetB7 proved to be the least accurate model, continuously demonstrating the lowest mean precision. Image rotation improves accuracy in certain circumstances, such as InceptionV3 and InceptionResNetV2, but decreases efficiency in others, such as DenseNet121 and EfficientNetB7.

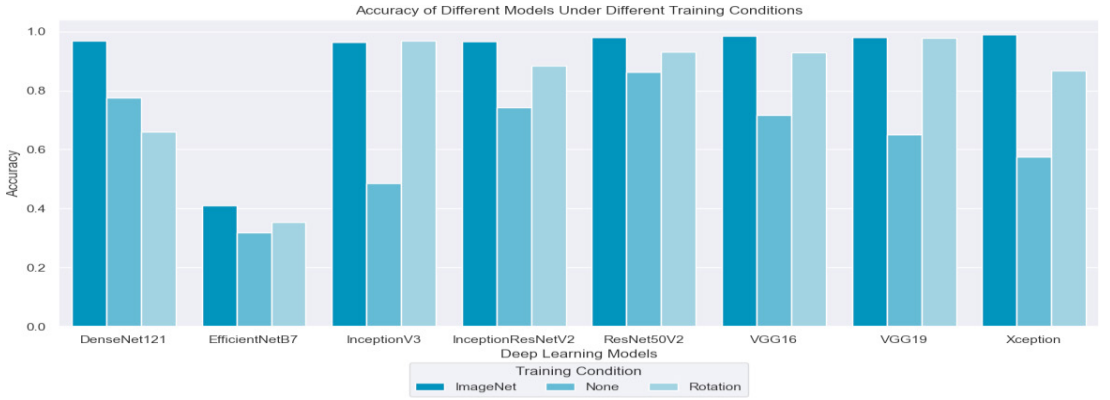


Fig. 5. Accuracy of Different Models Under Different Training Conditions.

4.2. Augmented Dataset

a) DL Framework Comparison for Brain Tumor Identification Using Augmented Dataset

InceptionResNetV2 consistently achieves the highest mean efficiency across all situations, making it the framework with the highest median precision performance. This implies that InceptionResNetV2 is well-suited to categorizing brain tumor pictures in the larger dataset. ResNet50V2, VGG16, and VGG19 operate as well, with consistently high mean precision across all classes. EfficientNetB7 seems to be the least accurate model across all classes, with the lowest mean accuracy on a consistent basis. The graph depicts the variation in reliability across every model and circumstance.

Models with minimal variability, such as VGG16 and VGG19, indicate more consistent effectiveness across all tumor classifications. Model Accuracy with Augmented Dataset is shown in figure 6.

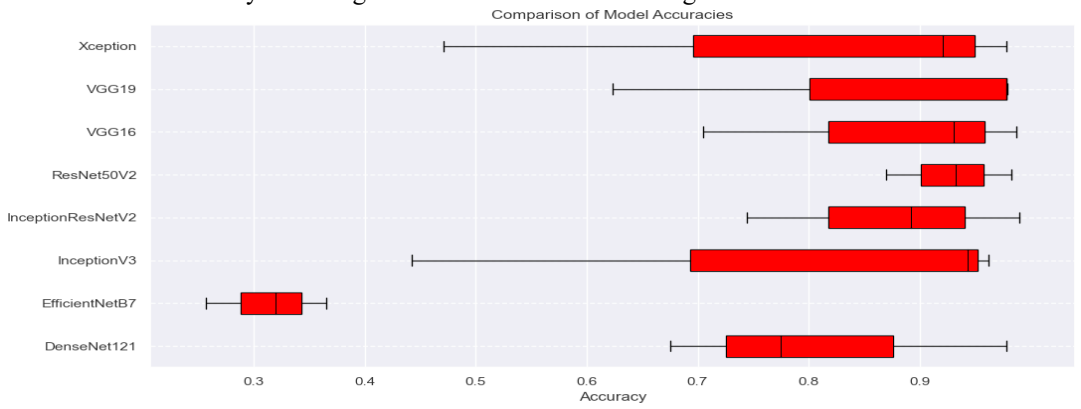


Fig. 6. Model Accuracy with Augmented Dataset.

b) Assessing the Effectiveness of DL Approaches for Classifying Brain Tumor

The InceptionResNetV2 (ImageNet) approach shows the best performance, with a fascinating precision of 0.990 when pretrained using ImageNet weights. VGG16, trained on ImageNet, performs well and consistently, with a precision of 0.987. The precision of VGG19 pretrained on ImageNet is appropriate at 0.978, while the accuracy of ResNet50V2 pretrained on ImageNet is excellent at 0.982. InceptionV3 achieves a high precision of 0.943 when trained with data enhancement (rotation), whereas Xception, trained using ImageNet, performs as well with a precision of 0.978. DenseNet121, which was trained on ImageNet, performs moderately well, with a precision of 0.978. Models such as ResNet50V2 with rotation-based data enhancement and Xception with rotation-based data enhancement show decent but reduced accuracies. Despite any pre-training, InceptionV3 has the lowest precision of 0.443, while EfficientNetB7 (none) has the least efficiency of 0.257. The top algorithms are InceptionResNetV2, VGG16, and VGG19 pretrained on ImageNet, whereas EfficientNetB7 and InceptionV3 without pre-training are the most desirable possibilities. Model Performance in Brain Tumor Classification is shown in figure 7.

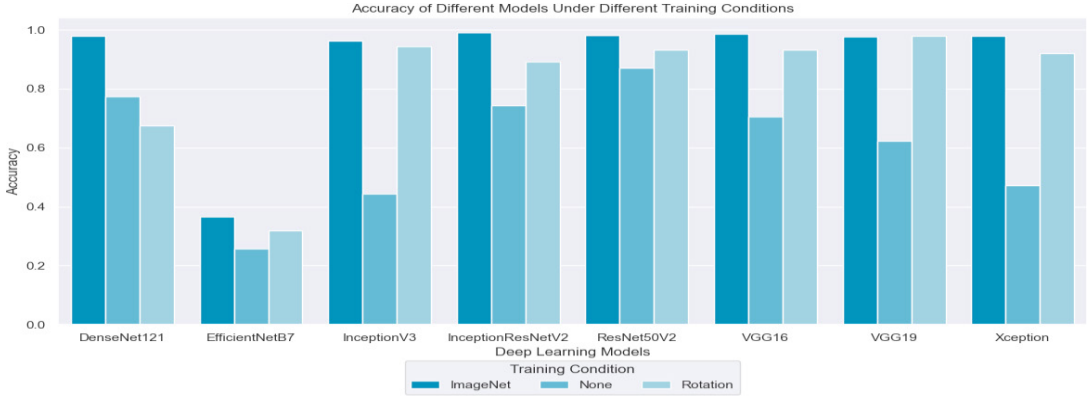


Fig. 7. Model Performance in Brain Tumor Classification.

5. Practical Implications

The innovative CNN model outperformed traditional approaches in the classification of brain tumors, achieving exceptional metrics, including 96.63% accuracy, 85.33% recall, 92.33% precision, and a 97.00% F1-score. The outstanding accuracy of the CNN model can be attributed to several key factors. Firstly, the model integrates robust mechanisms for detecting features, specifically designed to capture intricate patterns and features from brain MRI images associated with different tumor classes. The architecture, particularly the convolutional layers, facilitates the extraction of relevant and discriminative features crucial for precise classification.Furthermore, the CNN model demonstrates strong generalization capabilities through transfer learning from pre-trained models such as DenseNet121

and EfficientNet B7. The selection of model architectures is based on their proven efficacy in image classification tasks, contributing to the exceptional accuracy observed. Moreover, specialized architectures like InceptionResNetV2 and VGG19 exhibit a strong proficiency in identifying glioma tumors, highlighting the versatility of the approach. These findings suggest a comprehensive strategy that balances model architecture, feature extraction, and generalization, ultimately resulting in a highly accurate brain tumor classification system.

6. Originality/Value

6.1. Comparative Analysis of Brain Tumor Classification Models from augmented dataset

The unique CNN model outperformed all other brain tumor categorization approaches in our complete evaluation. As analyzed with the other algorithms, it had the greatest accuracy (96.63%), recall (85.33%), precision (92.33%), and F1-score (97.00%). This outstanding success may be attributed to the model's capacity to identify features from brain MRI images, its durability in dealing with image noise and unpredictability, and its amazing generalization capability. Other models performed admirably, with DenseNet121 and EfficientNet B7 exhibiting high accuracy rates, giving them viable options for brain tumor categorization. The accuracy of InceptionResNetV2 and VGG19 for glioma tumor detection was high. VGG19 and InceptionResNetV2 excelled in identifying glioma tumor with high precision. Although the Xception and ResNet50V2 models performed well, the Custom CNN, DenseNet121, and EfficientNet B7 models improved them marginally. Comparative Analysis of Brain Tumor Classification Models is shown in figure 8. Performance Comparison of Brain Tumor Classification Models is shown in table 5.

Table 5. Performance Comparison of Brain Tumor Classification Models.

Model Name	Metric	Glioma	Meningioma	No_Tumor	Pituitary
	Accuracy	0.9229	0.9229	0.9229	0.9229
Densenet121	Recall	0.7033	0.9867	0.9067	0.9752
	Precision	0.9769	0.6311	0.9963	0.7867
	F1-Score	0.8178	0.7698	0.9494	0.8708
EfficientNet B7	Accuracy	0.6713	0.6713	0.6713	0.6713
	Recall	0.8900	0.4800	0.0000	0.0000
	Precision	0.3944	0.2753	0.0000	0.0000
	F1-Score	0.5466	0.3499	0.0000	0.0000
	Accuracy	0.9479	0.9479	0.9479	0.9479
InceptionResNetV2	Recall	0.7633	0.9667	0.9233	0.9300
	Precision	0.9662	0.7379	0.9928	0.9588
	F1-Score	0.8529	0.8369	0.9568	0.9442
Inception_V3	Accuracy	0.9246	0.9246	0.9246	0.9246
	Recall	0.7400	0.8867	0.8133	0.9567
	Precision	0.9487	0.6982	1.0000	0.8416
	F1-Score	0.8315	0.7812	0.8971	0.8955
RestNet50V2	Accuracy	0.9408	0.9408	0.9408	0.9408
	Recall	0.6833	0.9400	0.9500	0.9533
	Precision	0.9809	0.7212	0.9661	0.9377
	F1-Score	0.8055	0.8162	0.9580	0.9455
VGG16	Accuracy	0.9567	0.9567	0.9567	0.9567
	Recall	0.7967	0.9533	0.9733	0.9300
	Precision	0.9598	0.7793	0.9832	0.9721
	F1-Score	0.8707	0.8576	0.9782	0.9506
VGG19	Accuracy	0.9508	0.9508	0.9508	0.9508
	Recall	0.8333	0.8567	0.9267	0.9900
	Precision	0.9058	0.8185	0.9929	0.9000
	F1-Score	0.8681	0.8371	0.9586	0.9429

Xception	Accuracy	0.9442	0.9442	0.9442	0.9442
	Recall	0.7233	0.9367	0.9500	0.9433
	Precision	0.9775	0.7454	0.9628	0.9279
	F1-Score	0.8314	0.8301	0.9564	0.9355
Custom_CNN	Accuracy	0.9663	0.9663	0.9663	0.9663
	Recall	0.8533	0.9233	0.9700	0.9833
	Precision	0.9808	0.8445	0.9898	0.9306
	F1-Score	0.9127	0.8822	0.9798	0.9562

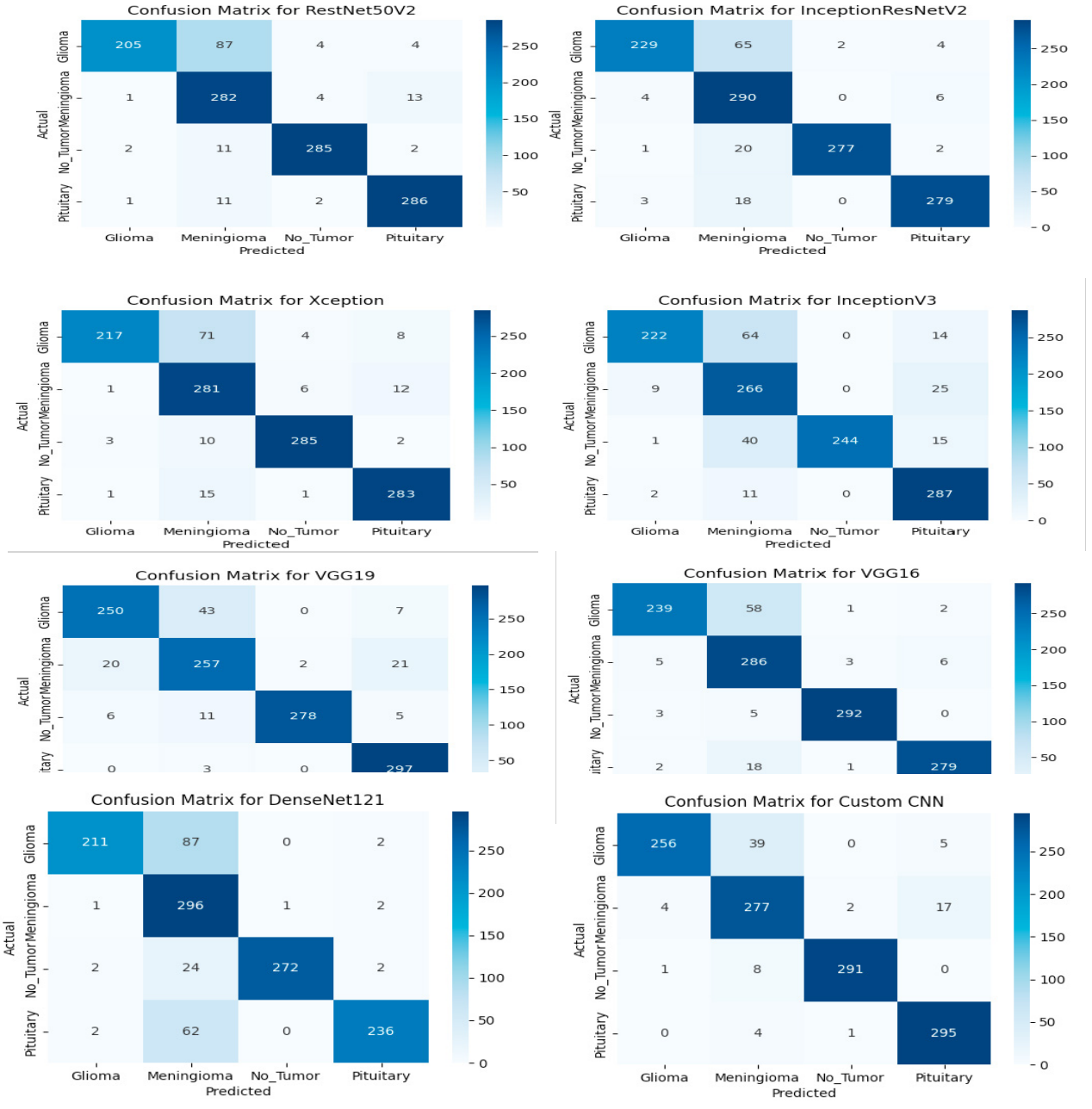


Fig. 8. Comparative Analysis of Brain Tumor Classification Models.

7. Conclusion & Future scope

In this research on brain tumor categorization, our study examined the efficiency of a wide range of different models other than the standard Convolutional Neural Network (CNN). Along with the CNN, extensively tested eight other models, consisting of Densenet121, EfficientNet B7, InceptionResNetV2, Inception_V3, RestNet50V2, VGG16, VGG19, and Xception. Despite our exclusive CNN model standing out as having the best performance, with a fantastic accurate rate of 96.63% and great recall, precision, and F1-score, the other models all demonstrated notable capabilities. For example, DenseNet121 and EfficientNet B7 have outstanding precision rates, resulting in viable choices for brain tumor categorization. VGG19 and InceptionResNetV2 displayed outstanding accuracy in detecting glioma tumor. Although algorithms such as Xception and RestNet50V2 fared effectively, our exclusive CNN, DenseNet121, and EfficientNet B7 models improved a bit, highlighting the possibility of continuing tuning. This study not only presents a cutting-edge CNN model but also provides vital insights into the various features of deep learning frameworks, with the potential to improve rapid brain tumor detection and patient treatment in the future. Future research should strive to modify and implement such models in real-world clinical settings to achieve measurable results.

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